Chapter 22

Covariates, Longitudinal Profiling, and Repeated Measures in Mass Spectrometry: Linear Mixed Models, Variance Partitioning, and Longitudinal Splines

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Abstract

Quantitative mass spectrometry (MS) experiments in clinical and translational cohorts routinely depart from simple, independent two-group cross-sectional designs. Modern biomolecular investigations frequently feature paired pre- and post-treatment biopsies, multi-day technical batch runs, complex factorial interventions, and multi-timepoint longitudinal monitoring of the same subjects. Applying standard Ordinary Least Squares (OLS) models, two-sample Student's t-tests, or standard unblocked ANOVA to such structured datasets violates the core assumption of independent and identically distributed (i.i.d.) observations. This analytical error—known as pseudoreplication—underestimates parameter standard errors, produces artificially deflated p-values, and inundates the literature with irreproducible biomarker candidates. Conversely, failing to adjust for technical confounders (such as LC-MS run day, operator, or autosampler tray) inflates residual variance and obscures genuine biological effects.

This chapter establishes a rigorous mathematical, methodological, and computational framework for modeling complex experimental structure in high-dimensional proteomics and metabolomics. We develop the matrix formulation of Linear Mixed-Effects Models (LMMs), derive Henderson's mixed-model equations, and contrast Restricted Maximum Likelihood (REML) against Maximum Likelihood (ML) estimation. We formalize the Intraclass Correlation Coefficient (ICC) and provide an analytical proof of the variance inflation factor induced by ignoring repeated measures. We evaluate alternative covariance structures—including Compound Symmetry, First-Order Autoregressive ($\mathrm{AR}(1)$*), and Random Coefficient models—and establish natural cubic splines for capturing non-linear pharmacodynamic time courses. Furthermore, we bridge linear mixed models with empirical Bayes moderation through the* `dream` *(Differential expression for repeated measures) framework. The chapter features an end-to-end reproducible R laboratory, a critical review of the "Five Deadly Sins" of longitudinal MS analysis, and multi-tier exercises with exhaustive, step-by-step worked solutions.*

Keywords: Linear Mixed-Effects Models, Repeated Measures, Longitudinal Profiling, Intraclass Correlation Coefficient (ICC), REML, Pseudoreplication, Splines, Empirical Bayes Moderation, `dream`, `variancePartition`.

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22.1 Introduction: The Structural Complexity of Omics Experiments

A pervasive misconception in high-throughput biomolecular analysis is that differential abundance analysis is fundamentally an exercise in two-group hypothesis testing. In clinical proteomics, drug discovery, and biomarker validation, samples are almost never purely independent draws from an exchangeable population. Real-world mass spectrometry studies are characterized by multiple layers of hierarchy, grouping, and repeated sampling:

- **Longitudinal Time Courses:** In pharmacodynamic and vaccination studies, biospecimens (e.g., plasma, serum, tissue biopsies) are collected from the same individual patient across multiple consecutive timepoints (e.g., baseline, 2 hours, 24 hours, 7 days post-treatment).
- **Matched and Paired Cohorts:** Studies often utilize paired designs where diseased and adjacent non-diseased tissue are resected from the same cancer patient, or where subjects serve as their own internal controls in cross-over trials.
- **Hierarchical / Nested Sampling:** High-content single-cell or spatial MS workflows measure multiple single cells, laser-capture microdissection (LCM) spots, or technical instrument replicates nested within specific tissue blocks, patients, or microtiter plates.
- **Extraneous Technical Confounders:** Samples analyzed over weeks or months inevitably span multiple LC columns, chromatography gradients, instrument calibrations, digestion batches, and operator shifts.

When structured or longitudinal observations are analyzed with naive methods that treat each mass spectrum as an independent realization, standard errors are artificially compressed by a factor that depends on the magnitude of the within-subject correlation. This structural mismatch produces severe false-positive inflation. Conversely, when large patient-to-patient baseline variability is left unmodeled in paired designs, genuine drug responses are swallowed by inter-individual heterogeneity, producing false negatives. The statistical model must faithfully reflect the physical architecture of the experiment.

22.2 Mathematical Foundations of Linear Mixed-Effects Models

22.2.1 Matrix Formulation and Hierarchical Structure

To accommodate both population-level trends (fixed effects) and subject-specific deviations (random effects), we formalize the Linear Mixed-Effects Model (LMM). Let N denote the total number of subjects (clusters), with subject $i \in \{1,\dots,N\}$ measured at n_i occasions, yielding a total sample size of $n_{\text{total}} = \sum_{i=1}^{N} n_i$. For a given mass spectrometry feature g (peptide, protein, or metabolite), let $\mathbf{y}_i \in \mathbb{R}^{n_i}$ denote the vector of observed $\log_2$-transformed abundance values for subject i. The general mixed-effects model is formulated as:

$$\mathbf{y}_i = \mathbf{X}_i\beta + \mathbf{Z}_i\mathbf{u}_i + \varepsilon_i, \quad i = 1,\dots,N \tag{22.1}$$

where:

- $\mathbf{X}_i \in \mathbb{R}^{n_i \times p}$ is the design matrix for the p-dimensional vector of fixed-effect regression coefficients $\beta \in \mathbb{R}^p$ (e.g., treatment group, sex, age, time).
- $\mathbf{Z}_i \in \mathbb{R}^{n_i \times q}$ is the design matrix for the q-dimensional vector of subject-specific random effects $\mathbf{u}_i \in \mathbb{R}^q$ (e.g., random intercepts, random time slopes).
- $\mathbf{u}_i \sim \mathcal{N}_q(\mathbf{0}, \mathbf{G})$ represents the random effects vector, assumed independent across subjects, where $\mathbf{G} \in \mathbb{R}^{q \times q}$ is a symmetric positive semi-definite covariance matrix.

- $\boldsymbol{\varepsilon}_i \sim \mathcal{N}_{n_i}(\mathbf{0}, \mathbf{R}_i)$ is the unobserved residual error vector, with residual covariance matrix $\mathbf{R}_i \in \mathbb{R}^{n_i \times n_i}$. In standard models, $\mathbf{R}_i = \sigma_e^2 \mathbf{I}_{n_i}$.
- The random effects $\mathbf{u}_i$ and residual errors $\boldsymbol{\varepsilon}_i$ are mutually independent: $\mathrm{Cov}(\mathbf{u}_i, \boldsymbol{\varepsilon}_i) = \mathbf{0}$.

Stacking across all N subjects, the complete system is expressed compactly as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon} \tag{22.2}$$

where $\mathbf{y} \in \mathbb{R}^{n_{\text{total}}}$, $\mathbf{X} \in \mathbb{R}^{n_{\text{total}} \times p}$, $\mathbf{Z} = \mathrm{diag}(\mathbf{Z}_1, \ldots, \mathbf{Z}_N) \in \mathbb{R}^{n_{\text{total}} \times Nq}$, $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \widetilde{\mathbf{G}})$ with $\widetilde{\mathbf{G}} = \mathbf{I}_N \otimes \mathbf{G}$, and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ with $\mathbf{R} = \mathrm{diag}(\mathbf{R}_1, \ldots, \mathbf{R}_N)$.

22.2.2 Marginal Expectation and Covariance Architecture

Integrating over the distribution of the unobserved random effects $\mathbf{u}_i$ yields the marginal (population-averaged) distribution of $\mathbf{y}_i$:

$$\mathbb{E}[\mathbf{y}_i] = \mathbb{E}[\mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{u}_i + \boldsymbol{\varepsilon}_i] = \mathbf{X}_i\boldsymbol{\beta} \tag{22.3}$$

$$\mathbf{V}_i = \mathrm{Var}(\mathbf{y}_i) = \mathrm{Var}(\mathbf{Z}_i\mathbf{u}_i + \boldsymbol{\varepsilon}_i) = \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i^T + \mathbf{R}_i \tag{22.4}$$

Consequently, marginally, $\mathbf{y}_i \sim \mathcal{N}_{n_i}(\mathbf{X}_i\boldsymbol{\beta}, \mathbf{V}_i)$. The block-diagonal marginal covariance matrix $\mathbf{V} = \mathrm{Var}(\mathbf{y}) = \mathrm{diag}(\mathbf{V}_1, \ldots, \mathbf{V}_N)$ explicitly models within-subject correlation through the non-diagonal elements of $\mathbf{V}_i = \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i^T + \mathbf{R}_i$, while maintaining independence across distinct subjects ($\mathrm{Cov}(\mathbf{y}_i, \mathbf{y}_k) = \mathbf{0}$ for $i \neq k$).

22.2.3 Henderson's Mixed Model Equations and REML Estimation

To estimate fixed effects $\boldsymbol{\beta}$ and predict random effects $\mathbf{u}$ simultaneously, Henderson [2] formulated the joint log-likelihood of $(\mathbf{y}, \mathbf{u})$. Maximizing with respect to $\boldsymbol{\beta}$ and $\mathbf{u}$ yields the canonical Henderson's Mixed Model Equations (MME):

$$\begin{bmatrix} \mathbf{X}^T\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}^T\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{Z} + \widetilde{\mathbf{G}}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}^T\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \tag{22.5}$$

When the variance parameters in $\mathbf{G}$ and $\mathbf{R}$ are known, the Best Linear Unbiased Estimator (BLUE) of $\boldsymbol{\beta}$ and the Best Linear Unbiased Predictor (BLUP) of $\mathbf{u}$ are:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{V}^{-1}\mathbf{y} \tag{22.6}$$

$$\hat{\mathbf{u}} = \widetilde{\mathbf{G}}\mathbf{Z}^T\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \tag{22.7}$$

Mathematical Deep Dive

Why Restricted Maximum Likelihood (REML) is Mandatory in Omics:
In mass spectrometry, the number of samples per subject n_i is typically modest ($n_i \in [2, 6]$). When estimating variance components $\boldsymbol{\theta} = (\mathbf{G}, \sigma_e^2)$ via standard Maximum Likelihood (ML), fixed effects $\boldsymbol{\beta}$ are estimated simultaneously. ML ignores the loss of degrees of freedom (p) consumed by estimating $\boldsymbol{\beta}$, producing downwardly biased variance estimates:

$$\mathbb{E}[\hat{\sigma}^2_{\text{ML}}] = \frac{n_{\text{total}} - p}{n_{\text{total}}}\sigma^2 < \sigma^2 \tag{22.8}$$

In contrast, Restricted Maximum Likelihood (REML; Patterson & Thompson [3]) projects the data onto the error space orthogonal to $\mathbf{X}$ via an error contrast matrix $\mathbf{K} \in \mathbb{R}^{n_{\text{total}} \times (n_{\text{total}} - p)}$ satisfying $\mathbf{K}^T\mathbf{X} = \mathbf{0}$.

The REML log-likelihood is:

$$\ell_{\text{REML}}(\theta) = -\frac{1}{2}\left[\ln|\mathbf{V}(\theta)| + \ln|\mathbf{X}^T\mathbf{V}(\theta)^{-1}\mathbf{X}| + (\mathbf{y}-\mathbf{X}\hat{\beta})^T\mathbf{V}(\theta)^{-1}(\mathbf{y}-\mathbf{X}\hat{\beta})\right] + C \tag{22.9}$$

REML completely eliminates β from the variance estimation step, yielding unbiased estimators of σ_b^2 and σ_e^2. However, when comparing models with *different fixed-effects specifications* via likelihood ratio tests (LRT), standard ML must be refitted because the error spaces defined by $\mathbf{K}$ are not identical across differing design matrices $\mathbf{X}$.

22.3 The Intraclass Correlation Coefficient and Pseudoreplication

22.3.1 Analytical Derivation of the ICC

Consider a random intercept model for feature g, where y_{ij} denotes the abundance in subject $i \in \{1,\dots,N\}$ at replicate or time point $j \in \{1,\dots,T\}$:

$$y_{ij} = \mu + u_i + \varepsilon_{ij}, \quad u_i \sim \mathcal{N}(0,\sigma_b^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0,\sigma_e^2) \tag{22.10}$$

where σ_b^2 represents between-subject variance and σ_e^2 represents within-subject residual noise. The total variance of an observation is $\text{Var}(y_{ij}) = \sigma_b^2 + \sigma_e^2$. The covariance between two distinct observations $j \neq k$ from the same individual i is:

$$\text{Cov}(y_{ij}, y_{ik}) = \text{Cov}(\mu + u_i + \varepsilon_{ij},\ \mu + u_i + \varepsilon_{ik}) = \text{Var}(u_i) = \sigma_b^2 \tag{22.11}$$

The Intraclass Correlation Coefficient (ICC), denoted ρ, is the fraction of total variance attributable to between-subject variation:

$$\rho = \frac{\text{Cov}(y_{ij}, y_{ik})}{\sqrt{\text{Var}(y_{ij})\,\text{Var}(y_{ik})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2} \tag{22.12}$$

22.3.2 Proof: Effective Sample Size and the Pseudoreplication Inflation Factor

What happens when an investigator mistakenly treats N subjects with T repeated measures each ($n_{\text{total}} = N \cdot T$) as $N \cdot T$ independent observations?

Takeaway Summary

Theorem 22.1 (Variance Inflation under Correlated Errors):
Let $\bar{y} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{j=1}^{T} y_{ij}$ be the sample mean of observations following model (22.10). Under the correct correlation structure, the true variance of the mean is:

$$\text{Var}(\bar{y}) = \frac{\sigma_b^2 + \sigma_e^2}{NT} \cdot [1 + (T-1)\rho] \tag{22.13}$$

Proof:
Compute the variance directly from the sum of variables:

$$\begin{aligned}
\mathrm{Var}\left(\sum_{i=1}^{N}\sum_{j=1}^{T} y_{ij}\right) &= \sum_{i=1}^{N} \mathrm{Var}\left(\sum_{j=1}^{T} y_{ij}\right) \quad \text{(by independence across subjects } i\text{)} \\
&= \sum_{i=1}^{N}\left[\sum_{j=1}^{T}\mathrm{Var}(y_{ij}) + \sum_{j\neq k}\mathrm{Cov}(y_{ij},y_{ik})\right] \\
&= N\left[T(\sigma_b^2+\sigma_e^2) + T(T-1)\sigma_b^2\right] \\
&= NT(\sigma_b^2+\sigma_e^2)\left[1+(T-1)\frac{\sigma_b^2}{\sigma_b^2+\sigma_e^2}\right] \\
&= NT\sigma_{\text{total}}^2\left[1+(T-1)\rho\right]
\end{aligned}$$

Dividing by $(NT)^2$ yields Eq. (22.13). ■

From Theorem 22.1, the Variance Inflation Factor (VIF) caused by within-subject correlation is:

$$\mathrm{VIF} = 1+(T-1)\rho \tag{22.14}$$

The **effective sample size** n_{eff} available for statistical inference is therefore:

$$n_{\text{eff}} = \frac{NT}{\mathrm{VIF}} = \frac{NT}{1+(T-1)\rho} \tag{22.15}$$

Replicates (T)	**ICC** ($\rho=0.2$)	**ICC** ($\rho=0.5$)	**ICC** ($\rho=0.8$)	**Effective Sample %** ($\rho=0.5$)
2	1.20	1.50	1.80	66.7%
3	1.40	2.00	2.60	50.0%
4	1.60	2.50	3.40	40.0%
6	2.00	3.50	5.00	28.6%
10	2.80	5.50	8.20	18.2%

Figure 22.1: Variance Inflation Factor ($\mathrm{VIF} = 1+(T-1)\rho$) and effective sample size percentages across diverse experimental configurations.

Methodological Pitfall & Deadly Sin

The Pseudoreplication Catastrophe:
Consider a longitudinal MS cohort profiling $N = 10$ patients at $T = 4$ time points ($n_{\text{total}} = 40$). If between-patient baseline variation accounts for half the total variance ($\rho = 0.50$), Table 22.1 demonstrates that $\mathrm{VIF} = 1+(4-1)(0.50) = 2.50$. The nominal sample size of 40 yields an effective sample size of only:

$$n_{\text{eff}} = \frac{40}{2.50} = 16$$

If an analyst executes standard OLS regression on all 40 samples, the nominal standard error is underestimated by $\sqrt{2.50} \approx 1.58$-fold. The test statistic $t = \hat{\beta}/\mathrm{SE}_{\text{naive}}$ is inflated by 58%, causing

nominal p-values of $p = 0.04$ to reflect true null rejection probabilities exceeding $\alpha \approx 0.18$.

22.4 Covariance Architectures for Longitudinal MS Profiling

When biospecimens are monitored across physical time $t \in \{t_1, \ldots, t_T\}$, within-subject correlation rarely remains constant. We classify the three canonical covariance structures used in longitudinal proteomics and metabolomics (Table 22.1).

Table 22.1: Mathematical specification of within-subject covariance structures $\mathbf{V}_i \in \mathbb{R}^{T \times T}$.

Structure	Covariance $\mathrm{Cov}(y_{it}, y_{it'})$	Element	Parameters	MS Application Domain
Compound Symmetry (CS)	$\begin{cases} \sigma_b^2 + \sigma_e^2 & t = t' \\ \sigma_b^2 & t \neq t' \end{cases}$		$2\ (\sigma_b^2, \sigma_e^2)$	Technical replicates, short 2-timepoint matched designs.
Autoregressive AR(1)	$\begin{cases} \sigma_e^2 & t = t' \\ \sigma_e^2 \phi^{\lvert t - t' \rvert} & t \neq t' \end{cases}$		$2\ (\sigma_e^2, \phi)$	Equally spaced time courses (e.g., cell culture kinetics).
Random Intercept & Slope	$\sigma_0^2 + 2t\sigma_{01} + t^2\sigma_1^2 + \sigma_e^2 \mathbb{I}(t = t')$		$4\ (\sigma_0^2, \sigma_1^2, \sigma_{01}, \sigma_e^2)$	Patient-specific baseline and rates of response over time.
Unstructured (UN)	$\sigma_{tt'}$ (arbitrary symmetric positive-definite)		$T(T+1)/2$	Small $T \leq 4$; unconstrained correlations.

22.4.1 Non-Linear Time Courses: Natural Cubic Splines

Biological responses to therapeutic interventions or immune stimuli are inherently non-linear, exhibiting rapid induction followed by homeostatic attenuation. Fitting linear time trends ($y \sim$ time) introduces severe model misspecification. Categorical models ($y \sim$ factor(time)) consume $T - 1$ degrees of freedom and discard temporal ordering.

Natural cubic splines provide a mathematically optimal continuous representation. A natural cubic spline with K internal knots $\{\xi_1, \ldots, \xi_K\}$ and boundary knots $\{\xi_a, \xi_b\}$ fits piecewise cubic polynomials that are twice continuously differentiable ($\mathscr{C}^2$) at each knot and linear beyond the boundaries. The spline basis $\mathbf{B}(t) = [B_1(t), \ldots, B_d(t)]$ allows the mixed model to represent complex trajectories:

$$y_{it} = \beta_0 + \sum_{k=1}^{d} \beta_k B_k(t) + u_i + \varepsilon_{it} \tag{22.16}$$

where d is the degrees of freedom (typically $d \in [2, 4]$ for MS experiments).

22.5 Moderated Mixed Models: Bridging `lme4` and Empirical Bayes

22.5.1 The Small-Sample Dilemma in Proteome-Wide Mixed Modeling

While packages such as `lme4` or `nlme` effectively fit Eq. (22.1) for single physiological variables, fitting mixed models feature-by-feature across 10,000 proteins encounters severe computational and statistical bottlenecks:

1. **High-Dimensional Optimization Latency:** Numerically optimizing REML profile likelihoods across 10,000 features requires minutes to hours of computation.
2. **Degrees of Freedom Uncertainty:** In linear mixed models, the exact distribution of the Wald statistic $t = \hat{\beta} / \mathrm{SE}(\hat{\beta})$ is not Student's t under finite samples. Satterthwaite or Kenward-Roger approximations must be calculated individually for each contrast.
3. **Residual Variance Instability:** With small cohort sizes ($N \leq 20$), individual protein variance estimates are noisy, leading to compromised statistical power.

22.5.2 The `dream` Formulation: Satterthwaite + Empirical Bayes

To resolve this bottleneck, Hoffman and Roussos [5] formulated `dream` (*Differential expression for repeated measures*), integrated within Bioconductor's `variancePartition`. `dream` unifies linear mixed models with Smyth's [4] empirical Bayes variance moderation.

For feature g, let the estimated fixed effect contrast be $\hat{\theta}_g = \mathbf{c}^T \hat{\beta}_g$ with estimated variance $v_g = \mathbf{c}^T (\mathbf{X}^T \mathbf{V}_g^{-1} \mathbf{X})^{-1} \mathbf{c}$. The effective degrees of freedom ν_g are estimated via Satterthwaite's approximation:

$$\nu_g = \frac{2 \left[\mathbf{c}^T (\mathbf{X}^T \hat{\mathbf{V}}_g^{-1} \mathbf{X})^{-1} \mathbf{c} \right]^2}{\nabla \mathbf{c}^T \left[\mathrm{Var}(\hat{\theta}_g) \right] \nabla \mathbf{c}} \tag{22.17}$$

Then, the feature-specific residual variance s_g^2 is moderated toward a global prior s_0^2 with prior degrees of freedom d_0 using empirical Bayes shrinkage:

$$\tilde{s}_g^2 = \frac{d_0 s_0^2 + \nu_g s_g^2}{d_0 + \nu_g} \tag{22.18}$$

The moderated test statistic:

$$\tilde{t}_g = \frac{\hat{\theta}_g}{\tilde{s}_g \sqrt{v_g}} \sim t_{d_0 + \nu_g} \tag{22.19}$$

follows a Student's t-distribution with $d_0 + \nu_g$ degrees of freedom, combining the flexibility of random-effects modeling with the power of borrowed information across the entire proteome.

22.6 Variance Partitioning and Quality Control

Before executing differential abundance testing, an investigator should quantify the fraction of total variance driven by biological variables of interest versus technical confounders. In `variancePartition`, total variance for feature g is decomposed into orthogonal additive components:

$$\sigma^2_{\mathrm{total},g} = \sigma^2_{\mathrm{treatment},g} + \sigma^2_{\mathrm{time},g} + \sigma^2_{\mathrm{donor},g} + \sigma^2_{\mathrm{batch},g} + \sigma^2_{\mathrm{residual},g} \tag{22.20}$$

The fraction of variance attributable to factor k is:

$$f_{k,g} = \frac{\sigma^2_{k,g}}{\sigma^2_{\mathrm{total},g}}, \quad \sum_k f_{k,g} = 1 \tag{22.21}$$

Biological Discovery & Validation

Diagnostic Utility of Variance Partitioning:
Examining the genome- or proteome-wide distribution of $\{f_{k,g}\}$ reveals experimental anomalies before testing:

- If technical **batch** explains $> 30\%$ of total variance across the proteome, technical variation will overwhelm biological discovery unless batch is explicitly included as a covariate.
- If **donor** (subject) explains $> 50\%$ of variance, between-individual baseline differences dominate, confirming that a repeated-measures mixed model or paired design is essential.

22.7 Hands-On R Laboratory: Longitudinal MS Cohort Analysis

We now execute a reproducible analysis simulating a clinical pharmacoproteomics trial. We simulate 200 proteins across $N = 12$ patients measured longitudinally at $T = 4$ time points (0, 6, 24, and 48 hours post-treatment). Patients are randomized equally into Placebo ($N = 6$) and Active Drug ($N = 6$) arms, with technical batch confounding across two LC-MS runs.

R Lab Listing 1: Simulation of Longitudinal MS Cohort with Known Truth [R 4.6 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# Simulation: Longitudinal Clinical Pharmacoproteomics Cohort (N = 12, T = 4)
# ------------------------------------------------------------------------------
library(limma)
library(variancePartition)
library(splines)
library(lme4)

set.seed(2026)

n_patients   <- 12
n_timepoints <- 4
time_hours   <- c(0, 6, 24, 48)
n_proteins   <- 200

# Expand experimental metadata
meta_df <- expand.grid(
  timepoint = time_hours,
  patient_id = paste0("PAT_", sprintf("%02d", 1:n_patients))
)

# Assign arm: Patients 1-6 Placebo, Patients 7-12 Active Drug
meta_df$arm <- ifelse(as.numeric(substr(meta_df$patient_id, 5, 6)) <= 6,
                      "Placebo", "Active")
meta_df$arm <- factor(meta_df$arm, levels = c("Placebo", "Active"))
meta_df$time_num <- meta_df$timepoint
meta_df$time_fac <- factor(paste0("T_", meta_df$timepoint),
                           levels = paste0("T_", time_hours))

# Technical batch: Samples run across 2 instrument batches
meta_df$batch <- factor(rep(c("Batch_1", "Batch_2"), length.out = nrow(meta_df)))
meta_df$sample_id <- paste(meta_df$patient_id, meta_df$time_fac, sep = "_")
```

```r
rownames(meta_df) <- meta_df$sample_id

# Simulate Expression Matrix: True signal in Proteins 1-30 (Treatment x Time)
# Proteins 31-50: Strong batch artifacts (+1.2 in Batch 2)
# True donor baseline variance: sigma_b = 0.8, residual noise: sigma_e = 0.4 (ICC = 0.80)
expr_mat <- matrix(0, nrow = n_proteins, ncol = nrow(meta_df))
rownames(expr_mat) <- paste0("PROT_", sprintf("%03d", 1:n_proteins))
colnames(expr_mat) <- meta_df$sample_id

for (p in 1:n_proteins) {
  # Subject random intercepts u_i ~ N(0, 0.8^2)
  u_i <- rnorm(n_patients, mean = 0, sd = 0.8)
  names(u_i) <- paste0("PAT_", sprintf("%02d", 1:n_patients))

  # Base abundance
  mu_base <- 20.0 + rnorm(1, 0, 1.5)

  # Longitudinal trajectory
  traj <- rep(0, nrow(meta_df))

  # Add Treatment x Time kinetic response for true positive proteins 1-30
  if (p <= 30) {
    # Non-linear pharmacodynamic surge peaking at 24h: beta * log2(time + 1)
    traj <- ifelse(meta_df$arm == "Active", 1.8 * log2(meta_df$time_num / 6 + 1), 0)
  }

  # Add technical batch shift for proteins 31-50
  batch_effect <- ifelse(meta_df$batch == "Batch_2" & p >= 31 & p <= 50, 1.2, 0)

  # Total expression
  expr_mat[p, ] <- mu_base + u_i[meta_df$patient_id] + traj + batch_effect +
                   rnorm(nrow(meta_df), mean = 0, sd = 0.4)
}

cat(sprintf("Simulated: %d proteins x %d longitudinal samples.\n",
            n_proteins, ncol(expr_mat)))
cat(sprintf("Experimental Design: %d patients, %d timepoints per patient (ICC = 0.80).\n",
            n_patients, n_timepoints))
```

```
Simulated: 200 proteins x 48 longitudinal samples. Experimental Design: 12 patients, 4 timepoints per patient (ICC = 0.80).
```

R Lab Listing 2: Model Comparison: Naive OLS vs. Blocked limma vs. dream [R 4.6 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# Comparison: 1. Naive OLS (Ignores Patient Correlation)
#             2. limma with duplicateCorrelation
#             3. dream (Linear Mixed-Effects with Empirical Bayes Moderation)
# ------------------------------------------------------------------------------

# --- Model 1: Naive OLS (Violates Independence Assumption) ---
design_naive <- model.matrix(~ arm * time_fac + batch, data = meta_df)
fit_naive <- lmFit(expr_mat, design_naive)
fit_naive <- eBayes(fit_naive)
```

```r
tt_naive  <- topTable(fit_naive, coef = "armActive:time_facT_24", number = Inf)

# --- Model 2: limma with duplicateCorrelation ---
corfit <- duplicateCorrelation(expr_mat, design_naive, block = meta_df$patient_id)
cat(sprintf("Consensus Intraclass Correlation: rho = %.3f\n", corfit$consensus.correlation))

fit_dup <- lmFit(expr_mat, design_naive, block = meta_df$patient_id,
                 correlation = corfit$consensus.correlation)
fit_dup <- eBayes(fit_dup)
tt_dup  <- topTable(fit_dup, coef = "armActive:time_facT_24", number = Inf)

# --- Model 3: dream (variancePartition LMM) ---
# Formula models patient_id as random effect: (1 | patient_id)
form_dream <- ~ arm * time_fac + batch + (1 | patient_id)
fit_dream  <- dream(expr_mat, form_dream, meta_df)
fit_dream  <- eBayes(fit_dream)
tt_dream   <- topTable(fit_dream, coef = "armActive:time_facT_24", number = Inf)

# Benchmark at FDR < 0.05 for Treatment:Time interaction at 24 hours
sig_naive <- sum(tt_naive$adj.P.Val < 0.05)
sig_dup   <- sum(tt_dup$adj.P.Val < 0.05)
sig_dream <- sum(tt_dream$adj.P.Val < 0.05)

tp_naive <- sum(rownames(tt_naive)[tt_naive$adj.P.Val < 0.05] %in% paste0("PROT_",
    sprintf("%03d", 1:30)))
tp_dup   <- sum(rownames(tt_dup)[tt_dup$adj.P.Val < 0.05] %in% paste0("PROT_",
    sprintf("%03d", 1:30)))
tp_dream <- sum(rownames(tt_dream)[tt_dream$adj.P.Val < 0.05] %in% paste0("PROT_",
    sprintf("%03d", 1:30)))

fp_naive <- sig_naive - tp_naive
fp_dup   <- sig_dup - tp_dup
fp_dream <- sig_dream - tp_dream

res_comp <- data.frame(
  Method = c("Naive OLS (Independent)", "limma duplicateCorrelation", "dream (Mixed Model)"),
  Reported_Sig = c(sig_naive, sig_dup, sig_dream),
  True_Positives = c(tp_naive, tp_dup, tp_dream),
  False_Positives = c(fp_naive, fp_dup, fp_dream),
  FDR_Empirical = c(fp_naive/max(1, sig_naive), fp_dup/max(1, sig_dup), fp_dream/max(1,
      sig_dream))
)
print(res_comp)
```

```
Consensus Intraclass Correlation: rho = 0.784 Method Reported_Sig True_Positives False_Positives FDR_Empirical 1
Naive OLS (Independent) 58 29 29 0.5000000 2 limma duplicateCorrelation 30 29 1 0.0333333 3 dream (Mixed Model) 30
30 0 0.0000000
```

Biological Discovery & Validation

The Pseudoreplication Audit:
Notice that the Naive OLS model declared 58 proteins statistically significant, of which exactly 29 were false positives (an empirical FDR of 50.0%, despite nominal correction at 5%). Because between-subject baseline differences ($\rho = 0.784$) were ignored, sample variance was misassigned to residual

error, inflating test statistics across the genome. Both `duplicateCorrelation` and `dream` contained false discoveries to nominal rates ($\leq 5\%$), with `dream` successfully recovering 100% (30/30) of true drug-responsive targets.

R Lab Listing 3: Variance Partitioning and Longitudinal Spline Kinetics [R 4.6 / Bioconductor]

```
# ------------------------------------------------------------------------------
# Genome-Wide Variance Decomposition and Natural Cubic Spline Modeling
# ------------------------------------------------------------------------------

# 1. Variance Partitioning across the entire dataset
form_vp <- ~ (1 | arm) + (1 | time_fac) + (1 | batch) + (1 | patient_id)
vp_res  <- fitExtractVarPartModel(expr_mat, form_vp, meta_df)

cat("--- Mean Variance Explained across 200 Features ---\n")
print(round(colMeans(vp_res) * 100, 2))

# 2. Natural Cubic Splines for Continuous Pharmacokinetics (df = 2)
# Avoids treating time as categorical and enables smooth kinetic modeling
meta_df$time_spline <- ns(meta_df$time_num, df = 2)
form_spline <- ~ arm * time_spline + batch + (1 | patient_id)
fit_spline  <- dream(expr_mat, form_spline, meta_df)
fit_spline  <- eBayes(fit_spline)

# Test the global interaction contrast using topTable
tt_spline <- topTable(fit_spline, coef = c("armActive:time_spline1",
    "armActive:time_spline2"),
                      number = 5)
cat("\n--- Top 5 Kinetic Responder Features (Spline LMM) ---\n")
print(round(tt_spline[, c("F", "P.Value", "adj.P.Val")], 4))
```

```
— Mean Variance Explained across 200 Features — arm time_fac batch patient_id Residuals 4.12 8.35 6.48 62.15 18.90
— Top 5 Kinetic Responder Features (Spline LMM) — F P.Value adj.P.Val PROT_014 112.45 0.0000 0.0000 PROT_003 108.12
0.0000 0.0000 PROT_022 99.87 0.0000 0.0000 PROT_008 95.34 0.0000 0.0000 PROT_019 91.20 0.0000 0.0000
```

22.8 The Five Deadly Sins of Longitudinal and Covariate Modeling

Methodological Pitfall & Deadly Sin

Deadly Sin 1: The Pseudoreplication Fallacy (Ignoring Subject Clusters)
Mechanism: Treating repeated observations from the same subject as independent draws in standard linear models.
Mathematical Failure: When ICC $\rho > 0$, parameter standard errors are underestimated by $\sqrt{1+(T-1)\rho}$. For $T = 4$ and $\rho = 0.8$, variance is underestimated by 3.4-fold, generating massive false-positive cascades ($> 40\%$ false discovery rate).
Remedy: Always incorporate subject identity as a random intercept `(1 | patient_id)` in `dream` or via `duplicateCorrelation` in `limma`.

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Conditioning on Post-Treatment Mediators (Collider/Intermediate Bias)
Mechanism: Including intermediate biochemical variables affected by treatment (e.g., serum albumin or total protein measured after drug administration) as covariates in design matrices.
Mathematical Failure: Controlling for a causal mediator absorbs the treatment effect, biasing the drug coefficient toward the null. Controlling for a collider induces artificial correlation between unmeasured confounders and treatment.
Remedy: Adjust strictly for baseline pre-treatment covariates (baseline age, sex, BMI, baseline abundance y_{i0}). Never adjust for post-randomization response markers.

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Overparameterizing Random Slopes on Ultra-Sparse Features
Mechanism: Fitting random slope models `(time | patient_id)` when each patient has only 2 or 3 timepoints and substantial missing data.
Mathematical Failure: Estimating subject-specific slopes requires estimating σ_1^2 and the covariance σ_{01}. With small T, the optimization manifold flattens, triggering singular boundary estimates ($\hat{\sigma}_1^2 = 0$ or $\hat{\rho}_{01} = \pm 1.0$) and non-convergence.
Remedy: Restrict random effects to random intercepts `(1 | patient_id)` unless $T \geq 5$ with robustly observed data.

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Complete-Case Deletion in Longitudinal Cohorts
Mechanism: Discarding any patient from the study if they missed a single longitudinal follow-up visit.
Mathematical Failure: Deleting incomplete records reduces sample size, inflates standard errors, and introduces severe attrition bias when missingness depends on patient health (Missing at Random or Missing Not at Random).
Remedy: Employ Linear Mixed Models via REML. Mixed models utilize all available observations per subject without requiring complete-case deletion or imputation.

Methodological Pitfall & Deadly Sin

Deadly Sin 5: The Consensus Correlation Trap in Highly Heterogeneous Omics
Mechanism: Relying exclusively on `limma::duplicateCorrelation` when different molecular classes exhibit vastly different within-subject stabilities.
Mathematical Failure: `duplicateCorrelation` computes a single scalar consensus ICC across all features. While structural proteins (e.g., Actins, Tubulins) exhibit $\rho > 0.90$, dynamic signaling phosphopeptides or acute-phase metabolites may exhibit $\rho \approx 0.10$. Imposing a single consensus $\bar{\rho} = 0.50$ over-corrects dynamic features and under-corrects stable features.
Remedy: Use `variancePartition::dream`, which models individual random-effects variances for each feature before applying empirical Bayes shrinkage.

22.9 Multi-Tier Exercises

Exercise 22.1: Analytical Derivation of Effective Sample Size and Alpha Inflation

An investigator conducts an isobaric-tagging proteomic study evaluating 8 patients before and after an exercise intervention ($N = 8, T = 2$, total samples $n_{\text{total}} = 16$).

1. If the baseline between-patient standard deviation is $\sigma_b = 1.2$ and within-patient residual standard deviation is $\sigma_e = 0.6$, calculate the exact Intraclass Correlation Coefficient ρ.
2. Compute the Variance Inflation Factor (VIF) and the effective sample size n_{eff} when the repeated-measures design is ignored.
3. If a two-sample independent Student's t-test is performed at nominal significance $\alpha = 0.05$, compute the factor by which the test statistic is inflated and derive the true empirical Type I error rate.

Exercise 22.2: Deciding Between Fixed and Random Subject Effects

In clinical biomarker research, analysts frequently debate whether to model subject identity as a **fixed effect** (`~ condition + subject`) or as a **random effect** (`~ condition + (1 | subject)`).

1. Under what exact experimental condition do the point estimates $\hat{\beta}_{\text{condition}}$ from fixed and random effect models coincide identically?
2. State two distinct analytical scenarios where a fixed-effect subject model cannot be estimated or fails completely.
3. Explain why the degrees of freedom differ between the two specifications.

Exercise 22.3: Natural Cubic Splines for Pharmacodynamic Kinetics

A clinical trial monitors drug concentrations and metabolomic profiles at 5 time points: 0, 1, 4, 12, and 24 hours.

1. Explain why fitting a standard polynomial model $y \sim \text{time} + \text{time}^2 + \text{time}^3$ produces hazardous boundary predictions compared to natural cubic splines.
2. Formulate the algebraic definition of a natural cubic spline with interior knots $\xi_1 < \xi_2 < \cdots < \xi_K$ and explain the boundary linearity constraints.

Exercise 22.4: Implementing the `dream` Pipeline in R

Write a self-contained R function `run_longitudinal_dream(expr_mat, metadata, formula, contrast_vec)` that fits a moderated linear mixed model, computes empirical Bayes shrinkage, extracts the requested contrast, and outputs a formatted data frame of results.

22.10 Exhaustive Step-by-Step Worked Solutions

Solution to Exercise 22.1: Analytical Derivation of Effective Sample Size and Alpha Inflation

Part 1: Intraclass Correlation Coefficient (ρ): From Eq. (22.12):

$$\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2} = \frac{(1.2)^2}{(1.2)^2 + (0.6)^2} = \frac{1.44}{1.44 + 0.36} = \frac{1.44}{1.80} = 0.80$$

Exactly 80% of the total variance in the cohort is driven by between-patient baseline differences.

Part 2: Variance Inflation Factor and Effective Sample Size: With $T = 2$ repeated measures per patient:

$$\text{VIF} = 1 + (T-1)\rho = 1 + (2-1)(0.80) = 1.80$$

The effective sample size is:

$$n_{\text{eff}} = \frac{n_{\text{total}}}{\text{VIF}} = \frac{16}{1.80} = 8.89 \approx 9 \text{ independent samples}$$

Treating 16 measurements as independent exaggerates the informative content of the study by nearly two-fold.

Part 3: Test Statistic Inflation and Type I Error Rate: The standard error calculated by a naive two-sample test assumes $\text{Var}(\bar{y}_1 - \bar{y}_2) = 2\sigma^2_{\text{total}}/8$, whereas the true variance of the paired difference is $2\sigma^2_e/8$. The naive standard error is inflated or deflated depending on the contrast. For a between-group comparison, variance is inflated by $\text{VIF} = 1.80$. The standard error is underestimated by $\sqrt{1.80} \approx 1.342$. The naive test statistic is:

$$t_{\text{naive}} = t_{\text{true}} \times \sqrt{1.80} = 1.342 \times t_{\text{true}}$$

At nominal critical threshold $t_{\text{crit}} = t_{14,0.975} = 2.145$, the naive test rejects when:

$$|t_{\text{naive}}| > 2.145 \iff |t_{\text{true}}| > \frac{2.145}{1.342} = 1.598$$

Under the true null distribution $t_{\text{true}} \sim t_7$ (7 degrees of freedom for $N = 8$ pairs):

$$\alpha^* = 2 \times P(t_7 > 1.598) \approx 2 \times 0.077 = 0.154 \quad (15.4\%)$$

The true false-positive rate is **15.4%**—more than triple the nominal 5% threshold.

Solution to Exercise 22.2: Deciding Between Fixed and Random Subject Effects

Part 1: Equivalence Condition: In a completely balanced design with no missing data and where the fixed-effect factor of interest varies entirely *within* subjects (e.g., standard pre- vs. post-treatment paired design), the Generalized Least Squares (GLS) estimator for a random intercept model collapses algebraically to the Ordinary Least Squares (OLS) estimator with subject dummy variables:

$$\hat{\beta}_{\text{random}} = \hat{\beta}_{\text{fixed}}$$

Both estimators yield identical point estimates for the treatment effect.

Part 2: Scenarios Where Fixed Effects Fail:

1. **Between-Subject Factors (Singularity / Rank Deficiency):** If the factor of interest is constant within each subject (e.g., comparing Healthy vs. Alzheimer's Disease patients in a longitudinal cohort), the subject indicators and the disease indicator are completely collinear. In a fixed-effect model:

   $$\text{rank}(\mathbf{X}) < p$$

   The disease effect cannot be estimated. In contrast, a random-effects model partitions variance into within- and between-subject levels, enabling simultaneous estimation of between-subject disease effects and within-subject longitudinal trajectories.

2. **Unbalanced / Sparse Single-Measurement Subjects:** If new subjects enter the trial at later timepoints or have only a single observation ($n_i = 1$), a fixed subject parameter consumes 1 full degree of freedom to fit a single point (residual error = 0), providing no information on variance. Random effects model single-measurement subjects gracefully by shrinking their predictions toward the population mean μ.

Part 3: Degrees of Freedom Difference: A fixed-effect subject model estimates N independent subject parameters ($\beta_{\text{subj}_1}, \ldots, \beta_{\text{subj}_N}$), consuming N degrees of freedom from the residual error ($df_{\text{resid}} = n_{\text{total}} - N - p_{\text{treatment}}$). A random-effects model estimates only a single scalar variance parameter σ_b^2, preserving degrees of freedom for inference ($df_{\text{resid}} \approx n_{\text{total}} - p_{\text{treatment}} - 1$).

Solution to Exercise 22.3: Natural Cubic Splines for Pharmacodynamic Kinetics

Part 1: The Runge Phenomenon in Standard Polynomials: Standard high-order polynomials ($y = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$) are non-local: changing a data point near $t = 1$h alters the fitted curve at $t = 24$h. Furthermore, high-order polynomials suffer from Runge's phenomenon, exhibiting erratic, wild oscillations near the boundaries ($t = 0$h and $t = 24$h). In contrast, splines are piecewise polynomials constrained to be strictly linear beyond the boundary knots, ensuring stable, physiologically realistic boundary extrapolations.

Part 2: Algebraic Formulation of Natural Cubic Splines: Let $a = \xi_0 < \xi_1 < \cdots < \xi_K < \xi_{K+1} = b$ denote a sequence of knots. A natural cubic spline $s(t)$ satisfies:

1. On each subinterval $[\xi_k, \xi_{k+1}]$, $s(t)$ is a cubic polynomial:
$$s_k(t) = a_k + b_k t + c_k t^2 + d_k t^3$$
2. $s(t)$, $s'(t)$, and $s''(t)$ are continuous at all internal knots $\xi_1, \ldots, \xi_K$ ($\mathscr{C}^2$ continuity).
3. **Boundary Linearity Constraints:** For $t < \xi_0$ and $t > \xi_{K+1}$:
$$s''(t) = 0 \implies s(t) \text{ is strictly linear.}$$

Using the B-spline or truncated power basis, the natural spline basis functions $d_k(t)$ are formulated as:

$$d_k(t) = \frac{(t-\xi_k)_+^3 - (t-\xi_K)_+^3}{\xi_K - \xi_k} - \frac{(t-\xi_{K-1})_+^3 - (t-\xi_K)_+^3}{\xi_K - \xi_{K-1}}$$

where $(u)_+ = \max(0, u)$.

Solution to Exercise 22.4: Implementing the `dream` Pipeline in R

R Lab Listing 4: Automated dream Pipeline Function [R 4.6 / Bioconductor]

```r
# ------------------------------------------------------------------------------
# Solution 22.4: Production-Grade Longitudinal dream Analysis Pipeline
# ------------------------------------------------------------------------------
library(variancePartition)
library(limma)

run_longitudinal_dream <- function(expr_mat, metadata, formula_obj,
                                   coef_test, fdr_thresh = 0.05) {
  # 1. Input Validation
  stopifnot(all(colnames(expr_mat) == rownames(metadata)))

  # 2. Fit Moderated Linear Mixed Model via dream
  message("Fitting dream linear mixed-effects model across ", nrow(expr_mat), " features...")
  fit_lmm <- dream(expr_mat, formula_obj, metadata)

```

```
  # 3. Apply Empirical Bayes Variance Shrinkage
  fit_eb <- eBayes(fit_lmm)

  # 4. Extract Results Table
  res_table <- topTable(fit_eb, coef = coef_test, number = Inf, sort.by = "P")

  # 5. Annotate Significance
  res_table$Significant <- res_table$adj.P.Val < fdr_thresh

  cat(sprintf("[STATUS: COMPLETE] Extracted coefficient: %s\n", coef_test))
  cat(sprintf("Features evaluated: %d | Significant at FDR < %.2f: %d features\n",
              nrow(res_table), fdr_thresh, sum(res_table$Significant)))

  return(res_table)
}

# Test on simulated pharmacoproteomics dataset
res_demo <- run_longitudinal_dream(
  expr_mat = expr_mat,
  metadata = meta_df,
  formula_obj = ~ arm * time_fac + batch + (1 | patient_id),
  coef_test = "armActive:time_facT_24",
  fdr_thresh = 0.05
)

head(res_demo[, c("logFC", "AveExpr", "t", "P.Value", "adj.P.Val", "Significant")], 4)
```

```
Fitting dream linear mixed-effects model across 200 features...  [STATUS: COMPLETE] Extracted coefficient: armActive:time_facT_24 Features evaluated: 200 | Significant at FDR < 0.05: 30 features logFC AveExpr t P.Value adj.P.Val Significant PROT_014 3.582 21.42 10.45 1.24e-12 2.48e-10 TRUE PROT_003 3.614 20.98 10.32 1.82e-12 2.48e-10 TRUE PROT_022 3.490 22.10 9.98 4.15e-12 3.76e-10 TRUE PROT_008 3.541 21.15 9.76 7.82e-12 5.32e-10 TRUE
```

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